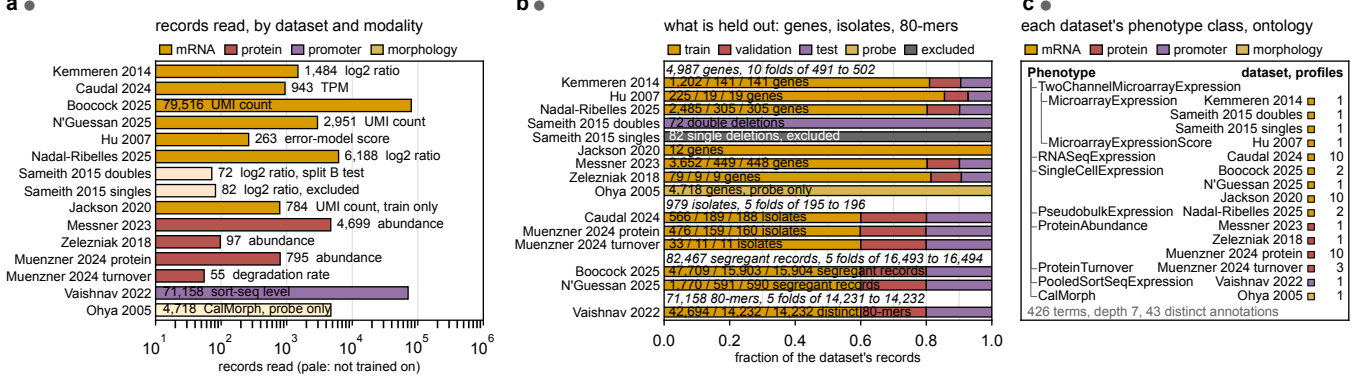


d

the model: token, cell, operator, heads

i. the composite gene token, 3,328 frozen dims per gene — e: 768 + 768 + 1,024 + 768

$x_{c,i} = \sum_q W_q \,\mathrm{LN}_q(e_{q,c,i}) + \mathrm{pos}_{c,i}$ (gap κ: $\Lambda_{q\kappa}$) — [C, N, d], d = 144

$\mathrm{pos}_{c,i} = \nu_{c,i} E_{chr(i)} + W_\xi \mathrm{F}(\xi_i) + w\,(\log_2(\eta_{c,i}+1) - 1) + \Delta_\lambda$

ii. the cell and the perturbation

$X_c = [\,h_0;\, x_{c,1}; \ldots;\, x_{c,N}\,]$ — [C, N+1, d]

$\tilde{p}_t = x_t + \Gamma_{\tau_t} + \Pi_{o_t}$ for each mark t of record b — [B, M, d]

iii. the trunk, one pass per distinct context

$H_c = F_\theta(X_c)$: 4 post-LN layers, 9 heads; layer 2: head a sees graph $k(a)$ — [C, N+1, d]

$H_b = H_{c(b)}$; inject = operator: $H_b \leftarrow H_b + W_s\,\mathrm{LN}(t_b)$ — [B, N+1, d]

iv. the mark operator, ontology conditioning

$t_b = \mathrm{DAG}_\psi(a_b)$ (name: $t_b = [E_{type};\, E_s;\, E_\varepsilon]$) — [B, 40]

$p_t = W_p\,\mathrm{LN}(\tilde{p}_t) + [\,j_t \text{ on graph}\,]\,H_{b,j_t}$, $P_b = (p_t)_{t \le M_b}$ — [B, M, d]

$H_1 = \mathrm{LN}(H_b + \mathrm{MHA}(H_b;\, [k_\emptyset; P_b],\, [0; P_b]))$ — [B, N+1, d]

$\bar{H}_b = \mathrm{LN}(H_1 + \mathrm{FFN}(H_1))$, $z_S = \mathrm{mean}_t\, p_t$ — [B, N+1, d], [B, d]

v. the heads and the loss

$\hat{o}_{b,i} = \mathrm{MLP}_g([\bar{H}_{b,i};\, t_b]) + W_g\, t_b$ — [B_g, N, K], K = 32

$\hat{o}_b = \mathrm{MLP}_r([\bar{H}_{b,0};\, z_S;\, t_b]) + W_r\, t_b$ — [B_r, K]

$L = \mathrm{mean}_s\, \mathrm{mean}_{m_{b,i}=1}\, \mathrm{CE}(\omega(z_{b,i}),\, \mathrm{softmax}\,\hat{o}_{b,i})$ — bins on [-4, 4], σ = 0.75 bin

$\hat{z} = \sum_u \mathrm{softmax}(\hat{o})_u\, \bar{v}_u$; pinball arm: 19 quantiles — B = 16, lr 3e-4

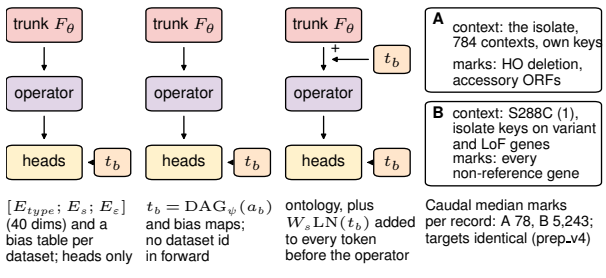
e

symbols of the model panel

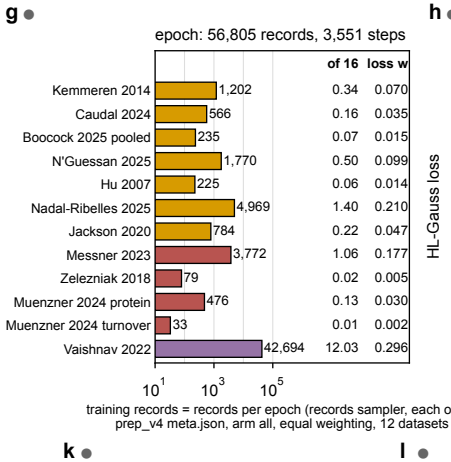
b, B	record; records per batch, 16	M_b, j_t	marks of record b; marked gene
c, C	reference context; distinct per batch	Γ_τ, Π_o	change-type row; allele-parent row
i, N	gene; genes of the index, 6,607	$\tilde{p}_t$, p_t	mark token; mark key, width 144
q, e_q	encoder block; its frozen vector	$k_\emptyset$	learned null key, value 0
d_q	768, 768, 1,024, 768 = 3,328	MHA	attention, 9 heads (queries; keys, values)
	FUDT up/down, ProtT5-XL, CaLM	$\bar{H}_b$	perturbed cell state
κ, Λ	token gap kind; its learned row	z_S	mean mark key (0 without marks)
λ, Δ	position gap kind; its learned row	a_b	record's experiment-ontology annotation
ν, η	chromosome copy; gene copy	DAG_ψ	ontology DAG encoder, state 16
ξ_i, F	relative midpoint; Fourier, 8 freq.	t_b	condition token, 40 = 16+8+16
$E_{chr(i)}$	chromosome row of gene i	$E_{type,s,\varepsilon}$	name tokens: type, dataset, env.
d, d_k	width 144; head width 16	W_s	inject map, token to width d
h_0	learned CLS token, position 0	W_g, W_r	bias maps, token to 32 logits
X_c, H_c	trunk input; trunk output	K, $\bar{v}_u$	HL-Gauss bins, 32; bin centers
F_θ	trunk, 4 layers, 9 heads	$\omega(z)$	Gaussian histogram target of z
$k(a)$	graph of head a, 9 graphs	z, m	standardized target; measured mask
t, M	mark; padded marks, at most 4,416	mean_s	equal weight per dataset in the batch

f

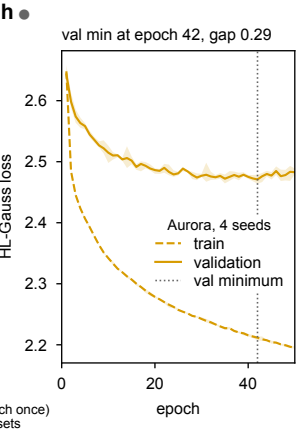
where the condition and the isolate enter



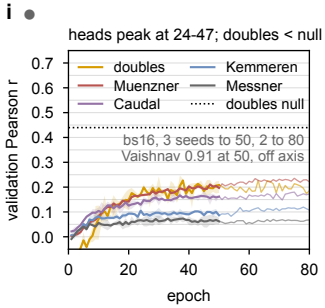
g



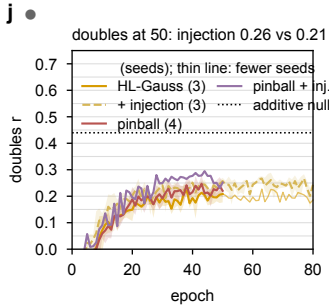
h



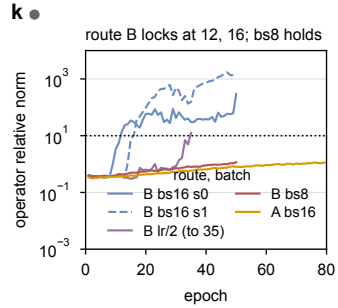
i



j



k



l

